

BJ NIGERIA

POST WELL REPORT FOR BILABRI D1

**PEAK PETROLEUM AND EQUATOR EXPLORATION NIGERIA LIMITED
PRELIMINARY STUDY
36" , 20" , 13 3/8" , 9 5/8" CASING & 7" LINER
OFFSHORE NIGERIA BLOCK OML 122
PORT HARCOURT – NIGERIA**

BJ SERVICES TECHNICAL REPRESENTATIVE

OLUWAGBEMIGA ILESANMI

NIGERIA

AUGUST, 2007



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PEAK PETROLEUM/ EQUATOR EXPLORATION.

PORT HARCOURT - NIGERIA

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BJ Services Company Ltd.

Client: Peak Petroleum & Equator Exploration Nigeria Limited

Project Name: Offshore Nigeria Block OML 122

Well Name: Bilabri D1

BJ Services Company Ltd.

COVER LETTER



August, 2007

Peak Petroleum Industries Nigeria Limited
And
Equator Exploration Nigeria Limited
Plot 1423, Amodu Tijani Street,
Off Sanusi Fafunwa Street,
Victoria Island, Lagos

RE: POST WELL CEMENTING REPORT ON BILABRI D1

Dear Sir,

Please find enclosed a summary of the cement operations performed by BJ Services on Peak Petroleum, Bilabri D1 in the month of November 2005 to January 2006.

The information presented here, is a summary of the events including the cementing technique, product usage, cost, highlights and lowlights from the cement jobs.

For this well, there were a lot of challenges however, we successfully cemented the entire strings in place without any operational crisis.

A lot of learning points have been gained from this particular operation and I feel confident that from our learning, BJ Services is positioned to meet and exceed your cementing expectations for the next well campaign.

We plan to continually meet and exceed Peak Petroleum expectations in quality service delivery.

BJ Services would like to thank you for your co-operation and team spirit shown during this operation and we are looking forward to your next well operation.

Sincere Regards
For **BJ Services Company Nigeria Limited.**

Oluwagbemiga Ilesanmi
Field Engineer.



BJ Services Company Ltd.
Client: Peak Petroleum/Equator Exploration Nigeria Ltd.
Project Name: Offshore Nigeria Block OML 122.
Well Name: Bilabri D1

INTRODUCTION TO WELL INFORMATION ACTUAL X PLANNED



INTRODUCTION TO WELL INFORMATION ACTUAL X PLANNED

Actual cementing performance data presented in this report represents information developed from actual job data cartridge simulated from the cement unit mini-monitor.

BJ Services' policy is to prepare pre and post job reports, this way we can be able to document the entire cementing operations and capture important learning points and areas of improvement during the cement job. These learning points are supposed to be carried over to subsequent jobs to achieve an improved quality cementing process.

For this well the Class of cement used throughout the well operation was Dyckerhoff Red Label Class "G", which is an API Monogram Class G Type HSR cement.

Well Overview

Prepared Date : August 2007
Location : Bulford Dolphin / Offshore Nigeria
Well : Bilabri – D1.

PLANNED WELL DATA

HOLE SIZE (inch)	CPPG. SIZE (inch)	CPPG. WT. (ppf)	MD-RKB (m)	TVD-RKB (m)	Est. BHST (deg. F)	Est. BHCT (deg. F)
36	30	310	238	238	68	59
26	20	129	438	438	91	84
17-1/2	13-3/8	72	1500	1500	133	108
12 1/4	9-5/8	47	3200	2982	270	203
8 1/4	7	26	4200	3836	225	163

ACTUAL WELL DATA

HOLE SIZE (inch)	CPPG. SIZE (inch)	CPPG. WT. (ppf)	MD-BRT (m)	TVD-BRT (m)	Est. BHST (deg. F)	Est. BHCT (deg. F)
36	30	310	250	250	68	59
26	20	129	438	438	91	84
17 – ½	13-3/8	72	1616	1616	133	108
12 – ¼	9-5/8	47	3200	2982	270	203
8 ¼	7	26	4000	3785	225	163

MUD DATA GIVEN:

HOLE SIZE (inch)	MUD TYPE	MUD WT. (ppg)	PV (cp)	YP (lb/100 ft2)
36	WBM	8.7	>80	>50
26	WBM	9.5	>80	>50
17.5	KCL MUD	9.5	>80	>50
12.5	OBM	13.5	>100	>65
8.25	OBM	14.5	>120	>75



BJ Services Company Ltd.
Client: Peak Petroleum/Equator Exploration Nigeria Ltd.
Project Name: Offshore Nigeria Block OML 122.
Well Name: Bilabri D1

DESIGN DETAILS ACTUAL X PLANNED



DESIGN DETAILS
ACTUAL VS PLANNED

20" CASING

	PLANNED		ACTUAL	
	LEAD	TAIL	LEAD	TAIL
<i>Cement type</i>	<i>API Class G</i>	<i>API Class G</i>	<i>API Class G</i>	<i>API Class G</i>
<i>Density (PPG)</i>	<i>12</i>	<i>15.8</i>	<i>12</i>	<i>15.8</i>
<i>Yield (ft³/sx)</i>	<i>2.517</i>	<i>1.17</i>	<i>2.517</i>	<i>1.17</i>
<i>Water type</i>	<i>Sea Water</i>	<i>Sea Water</i>	<i>Sea Water</i>	<i>Sea Water</i>
<i>Water required (gps)</i>	<i>14.6</i>	<i>5.05</i>	<i>14.6</i>	<i>5.05</i>
<i>Excess (%)</i>	<i>50</i>	<i>50</i>	<i>50</i>	<i>50</i>
<i>Estimated Slurry Volume (bbls)</i>	<i>433</i>	<i>112</i>	<i>405</i>	<i>152</i>
	ADDITIVES			
<i>FP-21L (gps)</i>	<i>0.04</i>	<i>0.04</i>	<i>0.04</i>	<i>0.04</i>
<i>A-3L (gps)</i>	<i>0.60</i>		<i>0.6</i>	
<i>A-7 (%bwoc)</i>		<i>0.05</i>		<i>0.04</i>
<i>BJ-2001</i>		<i>0.05</i>		<i>0.05</i>
	PROPERTIES			
<i>Fluid Loss Control (cc/30 min)</i>	<i>N/A</i>	<i>N/A</i>	<i>N/A</i>	<i>N/A</i>
<i>Thickening time(Hrs:Min to 40 Bc)</i>	<i>4:35</i>	<i>3:38</i>	<i>4:35</i>	<i>3:38</i>
<i>Thickening time(Hrs:Min to 70 Bc)</i>	<i>6:35</i>	<i>4:38</i>	<i>6:35</i>	<i>4:55</i>



13 3/8" CASING

	<i>PLANNED</i>		<i>ACTUAL</i>	
	<i>LEAD</i>	<i>TAIL</i>	<i>LEAD</i>	<i>TAIL</i>
<i>Cement type</i>	<i>API Class G</i>	<i>API Class G</i>	<i>API Class G</i>	<i>API Class G</i>
<i>Density (PPG)</i>	12.0	15.6	12.0	15.6
<i>Yield (ft³/sx)</i>	2.56	1.2	2.56	1.2
<i>Water type</i>	<i>Sea Water</i>	<i>Sea Water</i>	<i>Sea Water</i>	<i>Sea Water</i>
<i>Water required (gps)</i>	14.1	5.06	14.4	4.92
<i>Excess (%)</i>	30	0	30	0
<i>Estimated Slurry Volume (bbls)</i>	486	99	486	99
	<i>ADDITIVES</i>			
<i>FP-21L (gps)</i>	0.04	0.04	0.04	0.04
<i>R-21L (gps)</i>	0.032	0.032	0.027	0.043
<i>A-3L (gps)</i>	0.6		0.6	
<i>BJ-2001 (gps)</i>	0.5		0.5	
	<i>PROPERTIES</i>			
<i>Fluid Loss Control (cc/30 min)</i>	N/A	N/A	N/A	N/A
<i>Thickening time(Hrs:Min to 40 Bc)</i>	06:02	02:30	05:40	04:13
<i>Thickening time(Hrs:Min to 70 Bc)</i>	07:58	03:56	07:43	04:47



9 5/8" CASING

	<i>PLANNED</i>	<i>ACTUAL</i>
	<i>SINGLE SLURRY</i>	<i>SINGLE SLURRY</i>
<i>Cement type</i>	<i>API Class G</i>	<i>API Class G</i>
<i>Density (PPG)</i>	<i>15.0</i>	<i>15.8</i>
<i>Yield (ft³/sx)</i>	<i>1.73</i>	<i>1.73</i>
<i>Water type</i>	<i>Sea Water</i>	<i>Sea Water</i>
<i>Water required (gps)</i>	<i>5.85</i>	<i>5.85</i>
<i>Excess (%)</i>	<i>10</i>	<i>20</i>
<i>Estimated Slurry Volume (bbls)</i>	<i>530</i>	<i>160</i>
	<i>ADDITIVES</i>	
<i>FP-30L (gps)</i>	<i>0.06</i>	<i>0.06</i>
<i>BA-86L (gps)</i>	<i>1.5</i>	<i>1.5</i>
<i>BJ-2001 (gps)</i>	<i>0.25</i>	<i>0.25</i>
<i>CD-33L (gps)</i>	<i>0.2</i>	<i>0.2</i>
<i>R-21L (gps)</i>	<i>0.02</i>	<i>0.025</i>
<i>S-8 (%bwoc)</i>	<i>35</i>	<i>35</i>
	<i>PROPERTIES</i>	
<i>Fluid Loss Control (cc/30 min)</i>	<i>33</i>	<i>26</i>
<i>Thickening time(Hrs:Min to 40 Bc)</i>	<i>5:46</i>	<i>6:40</i>
<i>Thickening time(Hrs:Min to 70 Bc)</i>	<i>6:01</i>	<i>6:49</i>



7" LINER

	<i>PLANNED</i>	<i>ACTUAL</i>
	<i>SINGLE SLURRY</i>	<i>SINGLE SLURRY</i>
<i>Cement type</i>	<i>API Class G</i>	<i>API Class G</i>
<i>Density (PPG)</i>	<i>15.0</i>	<i>15.8</i>
<i>Yield (ft³/sx)</i>	<i>1.73</i>	<i>1.73</i>
<i>Water type</i>	<i>Sea Water</i>	<i>Sea Water</i>
<i>Water required (gps)</i>	<i>5.85</i>	<i>5.85</i>
<i>Excess (%)</i>	<i>20</i>	<i>20</i>
<i>Estimated Slurry Volume (bbls)</i>	<i>122</i>	<i>37</i>
	<i>ADDITIVES</i>	
<i>FP-30L (gps)</i>	<i>0.04</i>	<i>0.04</i>
<i>BA-86L (gps)</i>	<i>1.5</i>	<i>2</i>
<i>BJ-2001 (gps)</i>	<i>0.2</i>	<i>0.2</i>
<i>CD-33L (gps)</i>	<i>0.3</i>	<i>0.3</i>
<i>R-21L (gps)</i>	<i>0.027</i>	<i>0.005</i>
<i>S-8 (%bwoc)</i>	<i>35</i>	<i>35</i>
	<i>PROPERTIES</i>	
<i>Fluid Loss Control (cc/30 min)</i>	<i>41</i>	<i>23</i>
<i>Thickening time(Hrs:Min to 40 Bc)</i>	<i>5:23</i>	<i>3:47</i>
<i>Thickening time(Hrs:Min to 70 Bc)</i>	<i>5:01</i>	<i>3:55</i>



BJ Services Company Ltd.
Client: Peak Petroleum/Equator Exploration Nigeria Ltd.
Project Name: Offshore Nigeria Block OML 122.
Well Name: Bilabri D1

BJ SERVICES RECOMMENDED PRODUCTS



BJ SERVICES RECOMMENDED PRODUCTS

A-3L **Liquid Extender**
(Aqueous sodium silicate)

A clear fluid used in conjunction with sea water as an extender for cement slurries ranging in density from 11.0 - 15.3 ppg. A-3L tends to accelerate thickening time. Normally supplied in 55 US gallon drums or bulk.

A7-L **Liquid Accelerator**
(Calcium Chloride solution)

A thin clear fluid used as an accelerator to reduce thickening time and increase early compressive strength. 0.20 gal per sack is equivalent to 1.0 percent dry calcium chloride by weight of cement. Maximum concentration used 0.60 g/sk. Supplied in 55 US gallon drums, or bulk.

BJ-2001 **Multi-Purpose Additive**
(High advance polymer)

It was designed to provide ease of use, in terms of field mixing and handling. This product offers exceptional fluid loss control across a wide range of temperature and salinity conditions for primary and remedial cementing applications. It produces low free water slurries, can be extended to as low as 10ppg with LW-6/LW-7 or weighted to 21 PPG with W-10 or hematite. BJ-2001 by itself is able to control fluid loss, free water and rheology for fresh, seawater and light brines slurries. For Gas Control, BJ-2001 may be used either alone or combined with other cement matrix permeability modifiers (BA-100L or BA-58L) to produce extremely low fluid loss in a cost effective system. Package in 55 US gallon drums.

CD-33L **Liquid Dispersant**
(Sodium naphthalene sulphonate - sodium salt)

CD-33L is a thin, brown liquid, which is blended with mix water to lower the critical velocity for turbulent flow. Due to its thinning properties CD-33L can be used to prepare water-reduced slurries up to 17.5 ppg without the use of weighting materials. CD-33L also tends to slightly retard thickening time. Packaged in 55 US gallon drums.

FP-21L **Liquid Defoamer**

The use of defoamer reduces the amount of air entrainment during mixing, thus reducing mixing problems and loss of pump prime. FP-9L can be added to mix water or added directly to the cement mixing tub. FP-9L does not affect slurry properties, a minimum of 0.01 g/sk is generally recommended. Packaged in 55 US gallon drums.

MCS-G **Liquid Spacer Surfactant**
(Mixture of Non-Ionic surfactants)

MCS-G is an amber, low viscosity liquid surfactant possessing strong water wetting and mud thinning properties. Its incorporation in the MCS Spacer Systems produces a spacer fluid which is compatible with most types of drilling mud. Packaged in 55 US gallon drums.



BJ Services Company Ltd.
Client: Peak Petroleum/Equator Exploration Nigeria Ltd.
Project Name: Offshore Nigeria Block OML 122.
Well Name: Bilabri D1

LABORATORY TESTS – SLURRY RECIPES



CEMENT TEST REPORT

Project No.	13 3/8 CSG	Date	2-Dec-05
Requested by	Equator Petroleum	Rig number	Bulford Dolphin
Well number	Bilabri DX-1	Well Type	
Type of job/stage	13 3/8 CSG	Slurry type	LEAD

Title	Name	Signature	Date
Lab Analyst	Armstrong Odokpa		2-Dec-05
Supervisor	Ugochukwu Uchendu		2-Dec-05

WELL DATA

Description	Dia. (in.)	Weight	MD (m)	TVD (m)	BHST °F	BHCT °F
Previous Csg shoe						
Casing shoe			1500	1500	129	106
Stage Tool/liner top						
Open hole						

SLURRY COMPOSITION

ORDER OF MIXING	Products	Concentration	Units	Functions	Ref.
	FP-21L	0.04	gps	Antifoam	
	A-3L	0.60	gps	Extender	
	BJ-2001	0.500	gps	Multipurpose	
	R-21L	0.027	gps	Retarder	
	Dyckerhoff 'g' cmt.		sk	Cement	

Fresh H ₂ O	☐ Sea Water	☒ (gal/sk)	14.24	Mixing fluid (gal/sk)	15.40
Slurry Density (ppg)			12.0	Slurry Yield (cuft./sk)	2.54

CHLORIDES

Mix Water (mg/l)	
Mixing fluid (mg/l)	

RHEOLOGY

Fann (RPM)	300	200	100	60	30	6	3
After mixing							
At 80°F							
At 106°F	44	32	26			14	10

GEL STRENGTH

Gel 10 sec.	13
Gel 10 min.	35

FLUID PROPERTIES

Pv	27
Yp	17

FREE WATER

Temperature °F	Volume (ml)	%	Angle	Settling
106	0	0	0	nil

FLUID LOSS

Fluid loss API (cc/30 min.)	
HP/HT Fluid loss@106°F (cc/30min.)	

THICKENING TIME

		Initial	9Bc
Temperature °F	106	40 Bc	6hrs.02mins
Pressure (psi)	2900	70 Bc	7hrs.58mins
Time	29	100 Bc	8hrs.37mins

COMPRESSIVE STRENGTH

Temp.(°F) / Press(psi)	Time	UCA	Crush Cubes
129 / 3000	12 h	256 psi	psi
129 / 3000	24 h	401 psi	psi

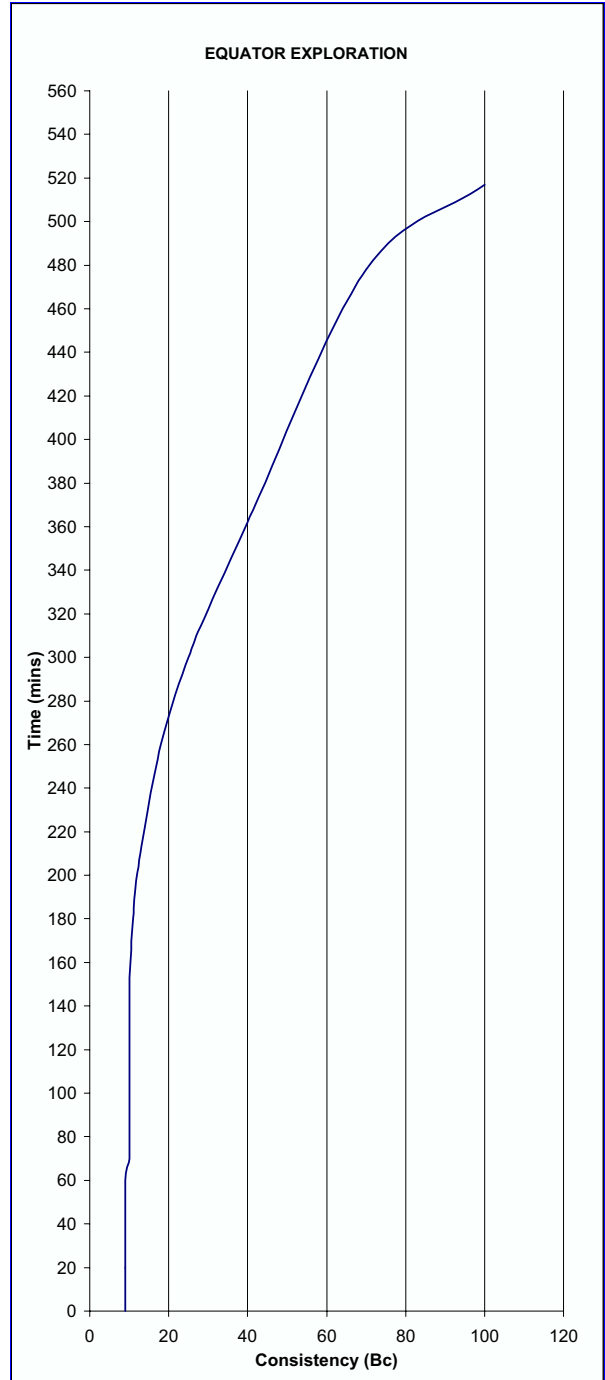
ADDITIVES/BATCH NUMBERS

FP-21L	040403169
A-3L	SD122100
BJ-2001	045
R-21L	488066

SPACER

Type	
Density	
600 rpm	
300 rpm	
200 rpm	
100 rpm	
6 rpm	
3 rpm	

CONSISTOMETER CHART



Remarks : Rig samples were used for these analysis.



CEMENT TEST REPORT

Project No.	13 3/8 CSG
Requested by	Equator Petroleum
Well number	Bilabri DX-1
Type of job/stage	13 3/8 CSG

Date	2-Dec-05
Rig number	Bulford Dolphin
Well Type	
Slurry type	TAIL

Title	Name	Signature	Date
Lab Analyst	Armstrong Odokpa		2-Dec-05
Supervisor	Ugochukwu Uchendu		2-Dec-05

WELL DATA

Description	Dia. (in.)	Weight	MD (m)	TVD (m)	BHST °F	BHCT °F
Previous Csg shoe						
Casing shoe			1500	1500	129	106
Stage Tool/liner top						
Open hole						

SLURRY COMPOSITION

ORDER OF MIXING	Products	Concentration	Units	Functions	Ref.
	FP-21L	0.04	gps	Antifoam	
	R-21L	0.043	gps	Retarder	
	Dyckerhoff 'g' cmt.		sk	Cement	

Fresh H ₂ O ☐ Sea Water ☒ (gal/sk)	5.06	Mixing fluid (gal/sk)	5.14
Slurry Density (ppg)	15.8	Slurry Yield (cuft./sk)	1.17

CHLORIDES

Mix Water (mg/l)	
Mixing fluid (mg/l)	

RHEOLOGY

Fann (RPM)	600	300	200	100	30	6	3
After mixing							
At 80°F							
At 106°F	115	77	66	52		18	12

GEL STRENGTH

Gel 10 sec.	13
Gel 10 min.	16

FLUID PROPERTIES

Pv	73.5
Yp	41.5

FREE WATER

Temperature °F	Volume (ml)	%	Angle	Settling
106	4	1.6	0	nil

FLUID LOSS

Fluid loss API (cc/30 min.)	
HP/HT Fluid loss@106°F (cc/30min.)	

THICKENING TIME

		Initial	12Bc
Temperature °F	106	40 Bc	4hrs.13mins
Pressure (psi)	2900	70 Bc	4hrs.47mins
Time	29	100 Bc	5hrs.13mins

COMPRESSIVE STRENGTH

Temp.(°F) / Press(psi)	Time	UCA	Crush Cubes
129 / 3000	12 h	890 psi	psi
129 / 3000	24 h	1879 psi	psi

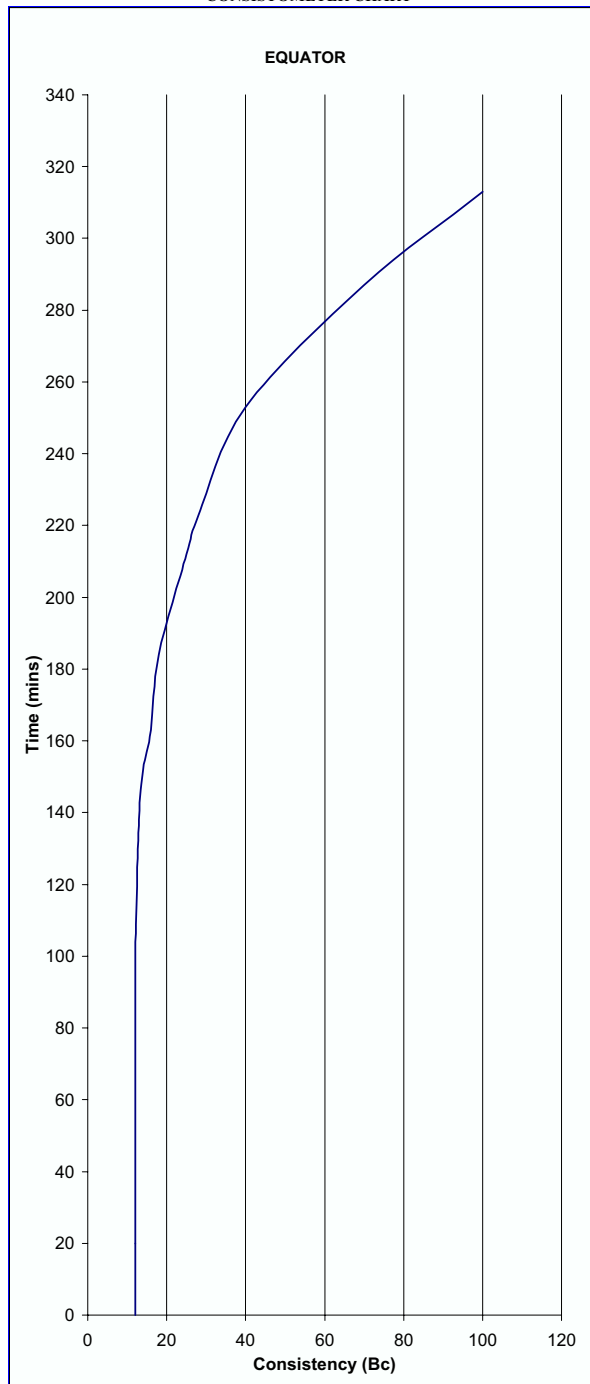
ADDITIVES/BATCH NUMBERS

FP-21L	040403169
R-21L	488066

SPACER

Type	
Density	
600 rpm	
300 rpm	
200 rpm	
100 rpm	
6 rpm	
3 rpm	

CONSISTOMETER CHART



Remarks : Rig samples were used for these analysis.



CEMENT TEST REPORT

Project No.	9 5.8" CSG
Requested by	Equator Petroleum
Well number	Bilabri Dx1
Type of job/stage	9 5.8" CSG

Date	21-Dec-05
Rig number	Bulford Dolphin
Well Type	
Slurry type	Single

Title	Name	Signature	Date
Lab Analyst	Armstrong Odokpa		21-Dec-05
Supervisor	Ugochukwu Uchendu		21-Dec-05

WELL DATA

Description	Dia. (in.)	Weight	MD (m)	TVD (m)	BHST °F	BHCT °F
Previous Csg shoe						
Casing shoe			3100	2840	213	154
Stage Tool/liner top						
Open hole						

SLURRY COMPOSITION

ORDER OF MIXING	Products	Concentration	Units	Functions	Ref.
	FP-30L	0.06	gps	Antifoam	
	BA-86L	1.50	gps	Bonding Agt	
	BJ-2001	0.25	gps	Multipurpose	
	CD-33L	0.20	gps	Dispersant	
	R-21L	0.025	gps	Retarder	
	S-8	35.00	% bwoc	Str. Retrogressn	
	Dyckerhoff 'g' cmt.		sk	Cement	

Fresh H ₂ O	☐ Sea Water	☒ (gal/sk)	5.85	Mixing fluid (gal/sk)	7.88
Slurry Density (ppg)		15.0	Slurry Yield (cuft./sk)	1.73	

CHLORIDES

Mix Water (mg/l)	
Mixing fluid (mg/l)	

RHEOLOGY

Fann (RPM)	300	200	100	60	30	6	3
After mixing							
At 80°F	79	54	30			5	3
At 154°F	52	37	22			4	2

GEL STRENGTH

Gel 10 sec.	3
Gel 10 min.	8

FLUID PROPERTIES

Pv	45
Yp	7

FREE WATER

Temperature °F	Volume (ml)	%	Angle	Settling
154	0.5	0.2	0	nil

FLUID LOSS

Fluid loss API (cc/30 min.)	
HP/HT Fluid loss@154°F (cc/30min.)	26

THICKENING TIME

		Initial	14Bc
Temperature °F	154	40 Bc	6hrs.23mins
Pressure (psi)	5800	70 Bc	6hrs39mins
Time	47	100 Bc	6hrs.44mins

COMPRESSIVE STRENGTH

Temp.(°F) / Press(psi)	Time	UCA	Crush Cubes
213 / 3000	12 h	psi	psi
213 / 3000	24 h	psi	psi

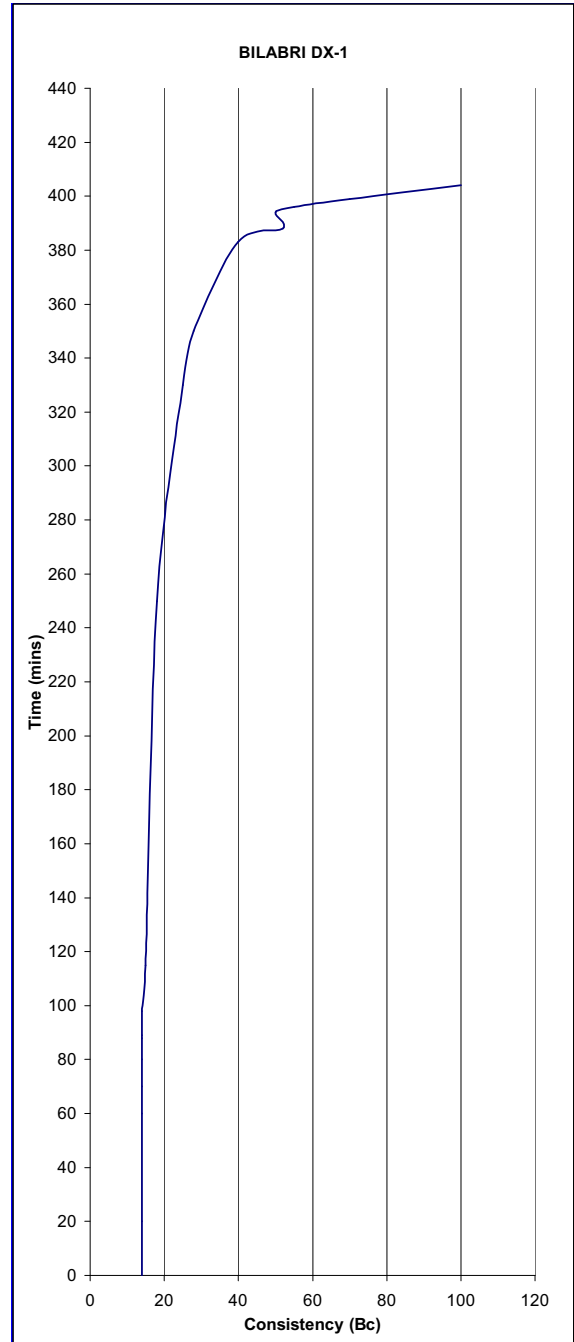
ADDITIVES/BATCH NUMBERS

FP-21L	040403169
BA-86L	018510229
BJ-2001	045
CD-33L	501213
R-21L	488066
S-8	4550069797

SPACER

Type	
Density	
600 rpm	
300 rpm	
200 rpm	
100 rpm	
6 rpm	
3 rpm	

CONSISTOMETER CHART



Remarks : Rig samples were used for these analysis.



CEMENT TEST REPORT

Project No.	9 5.8" CSG
Requested by	Equator Petroleum
Well number	Bilabri Dx1
Type of job/stage	9 5.8" CSG

Date	21-Dec-05
Rig number	Bulford Dolphin
Well Type	
Slurry type	Single

Title	Name	Signature	Date
Lab Analyst	Armstrong Odokpa		21-Dec-05
Supervisor	Ugochukwu Uchendu		21-Dec-05

WELL DATA

Description	Dia. (in.)	Weight	MD (m)	TVD (m)	BHST °F	BHCT °F
Previous Csg shoe						
Casing shoe			3100	2840	213	154
Stage Tool/liner top						
Open hole						

SLURRY COMPOSITION

ORDER OF MIXING	Products	Concentration	Units	Functions	Ref.
	FP-30L	0.06	gps	Antifoam	
	BA-86L	1.50	gps	Bonding Agt	
	BJ-2001	0.15	gps	Multipurpose	
	CD-33L	0.35	gps	Dispersant	
	R-21L	0.02	gps	Retarder	
	Dyckerhoff' g' cmt.		sk	Cement	

Fresh H ₂ O	☐ Sea Water	☒ (gal/sk)	4.21	Mixing fluid (gal/sk)	6.29
Slurry Density (ppg)			15.0	Slurry Yield (cuft./sk)	1.32

CHLORIDES

Mix Water (mg/l)	
Mixing fluid (mg/l)	

RHEOLOGY

Fann (RPM)	300	200	100	60	30	6	3
After mixing							
At 80°F	61	43	24			4	3
At 154°F	52	37	19			3	2

GEL STRENGTH

Gel 10 sec.	3
Gel 10 min.	18

FLUID PROPERTIES

Pv	49.5
Yp	2.5

FREE WATER

Temperature °F	Volume (ml)	%	Angle	Settling
154	0.5	0.2	0	nil

FLUID LOSS

Fluid loss API (cc/30 min.)	
HP/HT Fluid loss@154°F (cc/30min.)	46

THICKENING TIME

		Initial	14Bc
Temperature °F	154	40 Bc	6hrs.18mins
Pressure (psi)	5800	70 Bc	6hrs.36mins
Time	47	100 Bc	6hrs.40mins

COMPRESSIVE STRENGTH

Temp.(°F) / Press(psi)	Time	UCA	Crush Cubes
213 / 3000	12 h	psi	psi
213 / 3000	24 h	psi	psi

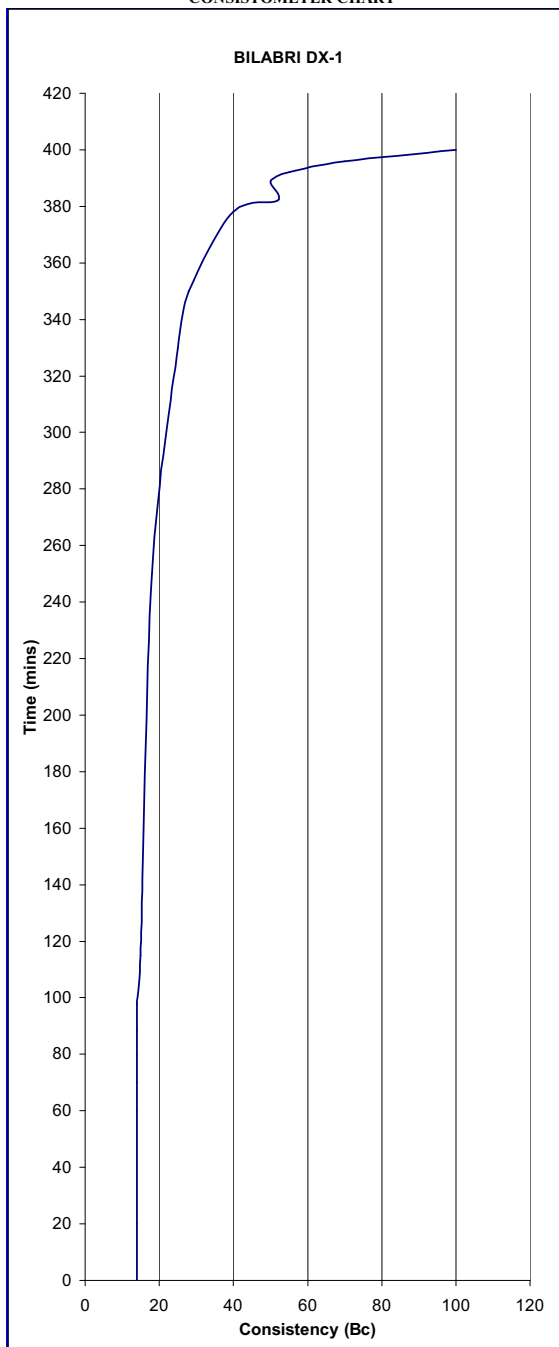
ADDITIVES/BATCH NUMBERS

FP-21L	040403169
BA-86L	018510229
BJ-2001	045
CD-33L	501213
R-21L	488066

SPACER

Type	
Density	
600 rpm	
300 rpm	
200 rpm	
100 rpm	
6 rpm	
3 rpm	

CONSISTOMETER CHART



Remarks : Rig samples were used for these analysis.



CEMENT TEST REPORT

Project No.	Squeeze
Requested by	Equator Petroleum
Well number	Bilabri Dx1
Type of job/stage	Squeeze @ 7" Shoe

Date	29-Jan-06
Rig number	Bulford Dolphin
Well Type	
Slurry type	Single

Title	Name	Signature	Date
Lab Analyst	Armstrong / Jude		29-Jan-06
Supervisor	Ugochukwu Uchendu		29-Jan-06

WELL DATA

Description	Dia. (in.)	Weight	MD (m)	TVD (m)	BHST °F	BHCT °F
Previous Csg shoe						
Casing shoe			3200	2977	225	191
Stage Tool/liner top						
Open hole						

SLURRY COMPOSITION

ORDER OF MIXING	Products	Concentration	Units	Functions	Ref.
	FP-21L	0.04	gps	Antifoam	
	R-21L	0.02	gps	Retarder	
	BJ-2001	0.30	gps	Multipurpose	
	CD-33L	0.45	gps	Dispersant	
	Dyckerhoff' g' cmt.		sk	Cement	

Fresh H ₂ O	☐ Sea Water	☒ (gal/sk)	4.44	Mixing fluid (gal/sk)	5.25
Slurry Density (ppg)			15.8	Slurry Yield (cuft/sk)	1.18

CHLORIDES

Mix Water (mg/l)	
Mixing fluid (mg/l)	

RHEOLOGY

Fann (RPM)	300	200	100	60	30	6	3
After mixing							
At 80°F	184	136	82			12	7
At 191°F	121	89	52			7	5

GEL STRENGTH

Gel 10 sec.	5
Gel 10 min.	19

FLUID PROPERTIES

Pv	103.5
Yp	17.5

FREE WATER

Temperature °F	Volume (ml)	%	Angle	Settling
191	0	0	0	nil

FLUID LOSS

Fluid loss API @ 191°F (cc/30 min.)	95
HP/HT Fluid loss @ 191°F (cc/30min.)	

THICKENING TIME

		Initial	14Bc
Temperature °F	191	40 Bc	3hrs.39mins
Pressure (psi)	7900	70 Bc	3hrs.49mins
Time	44	100 Bc	3hrs.52mins

COMPRESSIVE STRENGTH

Temp.(°F) / Press(psi)	Time	UCA	Crush Cubes
225 / 3000	12 h	psi	psi
225 / 3000	24 h	psi	psi

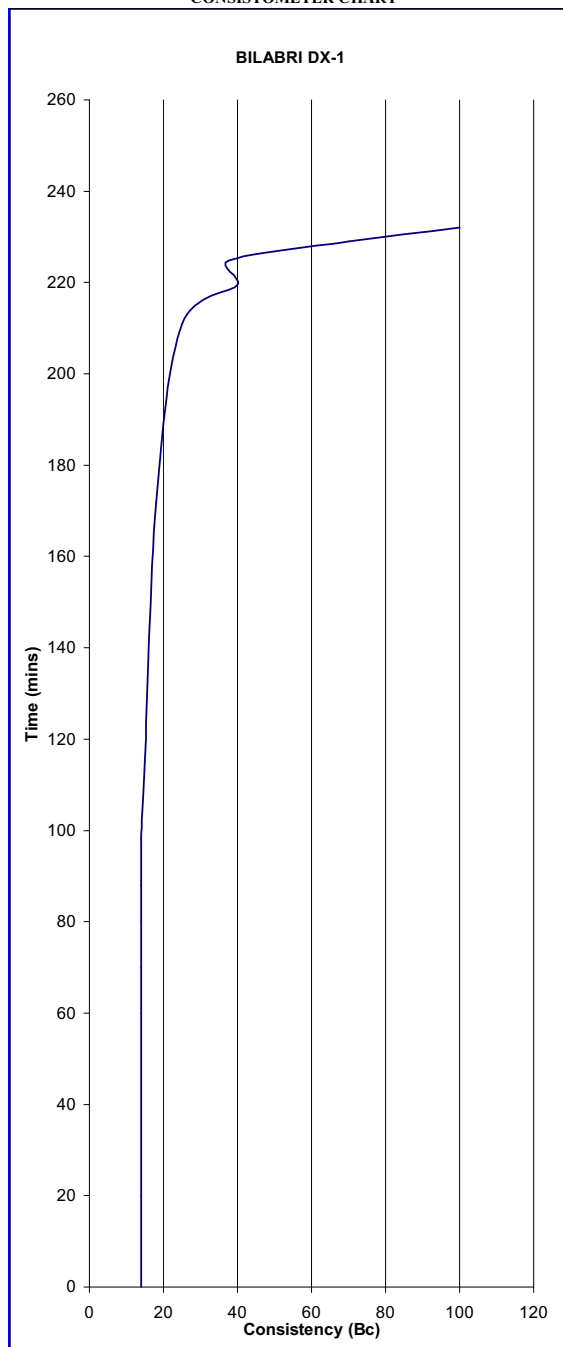
ADDITIVES/BATCH NUMBERS

FP-21L	040403169
R-21L	488066
BJ-2001	045
CD-33L	501213

SPACER

Type	
Density	
600 rpm	
300 rpm	
200 rpm	
100 rpm	
6 rpm	
3 rpm	

CONSISTOMETER CHART



Remarks : **Rig samples were used for these analysis.**



CEMENT TEST REPORT

Project No.	Squeeze
Requested by	Equator Petroleum
Well number	Bilabri Dx1
Type of job/stage	Squeeze @ 7" Shoe

Date	29-Jan-06
Rig number	Bulford Dolphin
Well Type	
Slurry type	Single

Title	Name	Signature	Date
Lab Analyst	Armstrong / Jude		29-Jan-06
Supervisor	Ugochukwu Uchendu		29-Jan-06

WELL DATA

Description	Dia. (in.)	Weight	MD (m)	TVD (m)	BHST °F	BHCT °F
Previous Csg shoe						
Casing shoe			3200	2977	225	191
Stage Tool/liner top						
Open hole						

SLURRY COMPOSITION

ORDER OF MIXING	Products	Concentration	Units	Functions	Ref.
	FP-21L	0.04	gps	Antifoam	
	R-21L	0.02	gps	Retarder	
	BJ-2001	0.30	gps	Multipurpose	
	CD-33L	0.45	gps	Dispersant	
	Dyckerhoff 'g' cmt.		sk	Cement	

Fresh H ₂ O	☐ Sea Water	☒ (gal/sk)	4.44	Mixing fluid (gal/sk)	5.25
Slurry Density (ppg)			15.8	Slurry Yield (cuft/sk)	1.18

CHLORIDES

Mix Water (mg/l)	
Mixing fluid (mg/l)	

RHEOLOGY

Fann (RPM)	300	200	100	60	30	6	3
After mixing							
At 80°F	184	136	82			12	7
At 191°F	121	89	52			7	5

GEL STRENGTH

Gel 10 sec.	5
Gel 10 min.	19

FLUID PROPERTIES

Pv	103.5
Yp	17.5

FREE WATER

Temperature °F	Volume (ml)	%	Angle	Settling
191	0	0	0	nil

FLUID LOSS

Fluid loss API @ 191°F (cc/30 min.)	95
HP/HT Fluid loss @ 191°F (cc/30 min.)	

THICKENING TIME

		Initial	14Bc
Temperature °F	191	40 Bc	3hrs.39mins
Pressure (psi)	7900	70 Bc	3hrs.49mins
Time	44	100 Bc	3hrs.52mins

COMPRESSIVE STRENGTH

Temp.(°F) / Press(psi)	Time	UCA	Crush Cubes
225 / 3000	12 h	psi	psi
225 / 3000	24 h	psi	psi

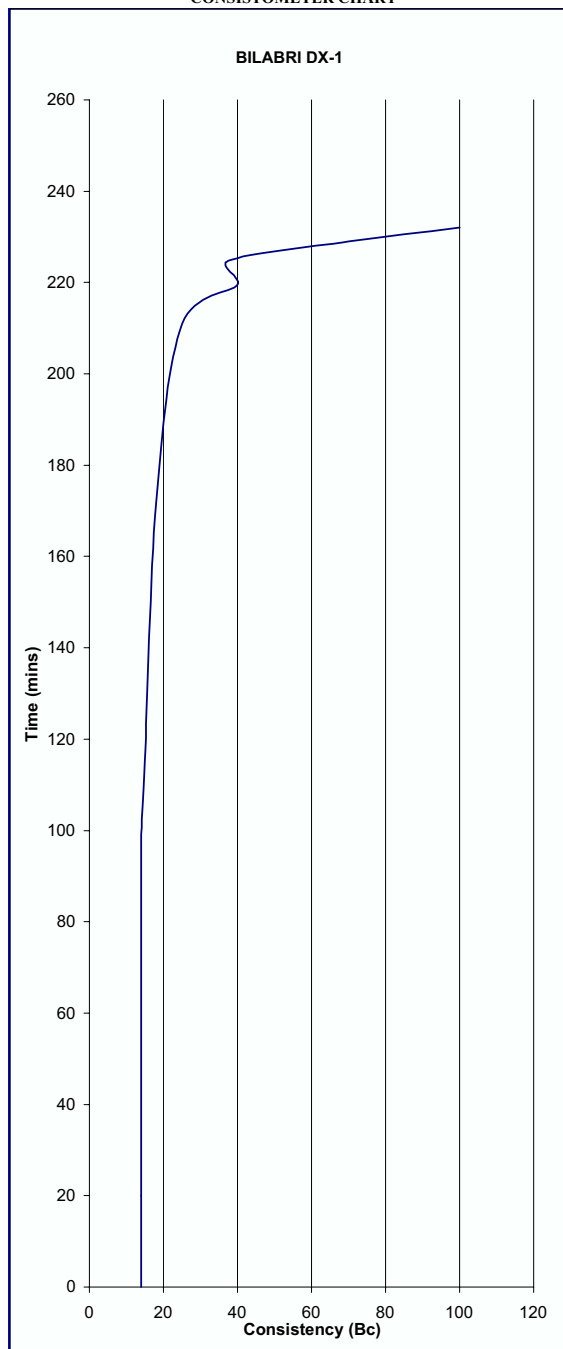
ADDITIVES/BATCH NUMBERS

FP-21L	040403169
R-21L	488066
BJ-2001	045
CD-33L	501213

SPACER

Type	
Density	
600 rpm	
300 rpm	
200 rpm	
100 rpm	
6 rpm	
3 rpm	

CONSISTOMETER CHART



Remarks : **Rig samples were used for these analysis.**



CEMENT TEST REPORT

Project No.	Squeeze 2	Date	18-Feb-06
Requested by	Equator Petroleum	Rig number	Bulford Dolphin
Well number	Bilabri DX-1	Well Type	
Type of job/stage	Squeeze	Slurry type	Single Slurry

Title	Name	Signature	Date
Lab Analyst	Armstrong Odokpa		18-Feb-06
Supervisor	Ugochukwu Uchendu		18-Feb-06

WELL DATA

Description	Dia. (in.)	Weight	MD (m)	TVD (m)	BHST °F	BHCT °F
Previous Csg shoe						
Casing shoe			3600	3250	270	230
Stage Tool/liner top						
Open hole						

SLURRY COMPOSITION

ORDER OF MIXING	Products	Concentration	Units	Functions	Ref.
	FP-30L	0.04	gps	Antifoam	
	BA-86L	2.00	gps	Bonding Agt	
	FL-45LN	0.15	gps	Fluid-loss Addit.	
	CD-33L	0.30	gps	Dispresant	
	R-14L	0.045	gps	Retarder	
	S-8	35.00	sk	Str. Retrogressn	
	Dyckerhoff "G" cement		sk	Cement	

Fresh H ₂ O	☐ Sea Water	☒ (gal/sk)	3.9	Mixing fluid (gal/sk)	6.44
Slurry Density (ppg)			15.8	Slurry Yield (cuft./sk)	1.54

CHLORIDES

Mix Water (mg/l)	
Mixing fluid (mg/l)	

RHEOLOGY

Fann (RPM)	300	200	100	60	30	6	3
After mixing							
At 80°F	54	36	20			4	2
At 230°F	78	57	46			6	4

GEL STRENGTH

Gel 10 sec.	5
Gel 10 min.	8

FLUID PROPERTIES

Pv	48
Yp	30

FREE WATER

Temperature °F	Volume (ml)	%	Angle	Settling
230	0	0	0	nil

FLUID LOSS

Fluid loss API @230°F(cc/30 min.)	28
HP/HT Fluid loss@230°F (cc/30min.)	

THICKENING TIME

		Initial	12Bc
Temperature °F	230	40 Bc	5hrs.10mins
Pressure (psi)	8700	70 Bc	7hrs.06mins
Time	47	100 Bc	7hrs.54mins

COMPRESSIVE STRENGTH

Temp.(°F) / Press(psi)	Time	UCA	Crush Cubes
270 / 3000	12 h	psi	psi
270 / 3000	24 h	psi	psi

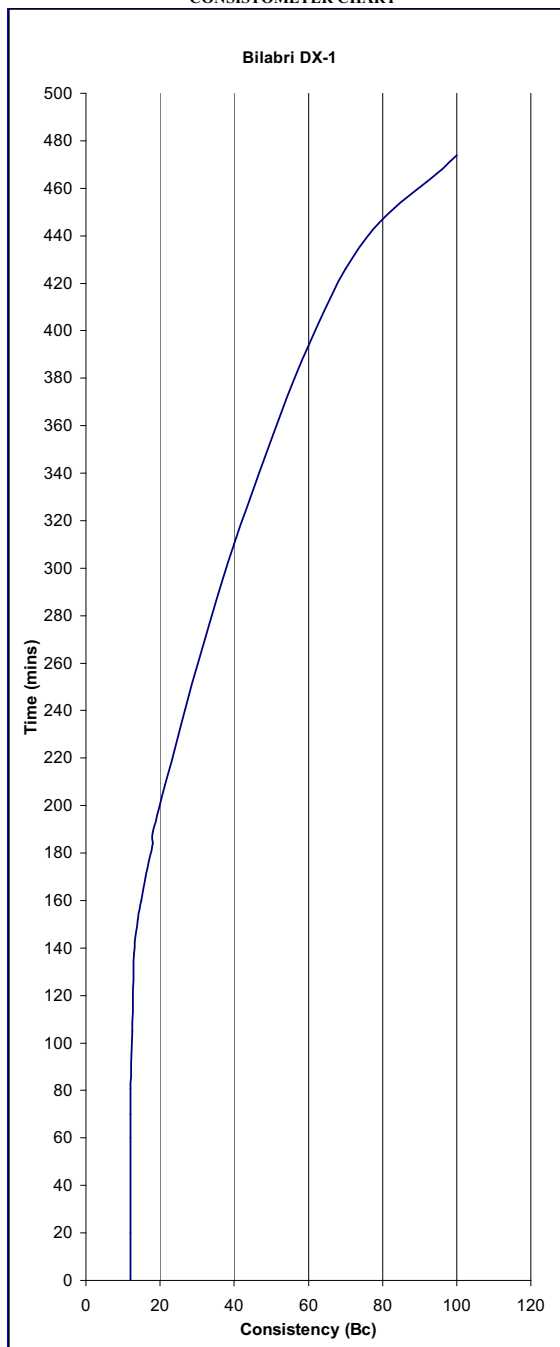
ADDITIVES/BATCH NUMBERS

FP-30L	158FZ
BA-86L	018510229
FL-45L	1005
CD-33L	501213
R-14L	2106
S-8	4550069797

SPACER

Type	
Density	
600 rpm	
300 rpm	
200 rpm	
100 rpm	
6 rpm	
3 rpm	

CONSISTOMETER CHART



Remarks : Rig samples were used for these analysis.



BJ Services Company Ltd.

Client: Peak Petroleum & Equator Exploration Nigeria Limited

Project Name: Offshore Nigeria Block OML 122

Well Name: Bilabri D1

BJ Services Company Ltd.

CEMENT JOB REPORTS



CEMENT JOB REPORT

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CUSTOMER EQUATOR PETROLEUM		DATE 28/11/05		F.R. #		SERV. SUPV. KENE SIMMON	
LE & WELL NAME - OCSG BILAGRI DX1		LOCATION				COUNTY-PARISH-BLOCK	
DISTRICT NIGERIA		DRILLING CONTRACTOR RIG # BULFORD DOLPHIN				TYPE OF JOB 30" CASING	
SIZE & TYPE OF PLUGS		LIST-CSG-HARDWARE		PHYSICAL SLURRY PROPERTIES			
TOP				SLURRY WGT PPG	SLURRY YLD FT ³	WATER GPS	PUMP TIME HR:MIN
BTM							
MATERIALS FURNISHED BY BJ				13.5	1.786	8.99	2.40
CLASS 'G' CEMENT, A7, FP-2IL, A-3L.				15.8	1.174	5.09	0.47
Available Mix Water _____ Bbl.				Available Displ. Fluid _____ Bbl.		TOTAL 396	
HOLE		TBG-CSG-D.P.		TBG-CSG-D. P.		COLLAR DEPTHS	
% EXCESS	DEPTH	SIZE	WGT.	TYPE	DEPTH	SIZE	WGT.
36"	100°	250m	30"	30#	240m		
LAST CASING		PKR-CMT RET-BR PL-LINER		PERF. DEPTH		TOP CONN	
SIZE	WGT.	TYPE	DEPTH	BRAND & TYPE	DEPTH	TOP	BTM
CAL. DISPL. VOL. - Bbl		CAL PSI		CAL MAX PSI		OP. MAX	
TBG.	CSG.	CSG/D.P.	TOTAL	BUMP PLUG	TO REV.	SQ. PSI	RATED
COMMENTS: TROUBLE SETTING TOOL, RUNNING CSG, ETC. PRIOR TO CEMENTING: _____							
PRESSURE/RATE DETAIL				JOB EVENT SUMMARY			
TIME HR:MIN.	PRESSURE-PSI		RATE BPM	Bbl. FLUID PUMPED	FLUID TYPE	SAFETY MEETING: BJ CREW ☐ CO. REP. ☐	
	PIPE	ANNULUS				TEST LINES PSI	
07:30						CIRCULATING WELL RIG ☐ BJ ☐	
07:35	1000					FLUSH LINE	
07:45			68bpm	20Bbl	SEA H2O	PRESSURE TEST LINE	
08:05	110		38bpm	254	SLURRY	WATER AHEAD	
08:20	120		3	142	SLURRY	MIX AND PUMP 13.5 PPG SLURRY	
08:19	230		4	54	SEA H2O	MIX AND PUMP 15.8 PPG SLURRY	
						DISPLACEMENT	
BUMPED PLUG							
Y	N	PSI TO BUMP PLUG	TEST FLOAT EQUIP.	BARRELS CEMENT RETURNED	TOTAL Bbl. PUMPED	PSI LEFT ON CSG	CAL. TOP OF CEMENT
							SERV. SUPV. KENE / SIMMON



CEMENT JOB REPORT

Page ____ of ____

CUSTOMER AR/EQUATOR PETROLEUM	DATE 04/12/05	F.R. #	SERV. SUPV. KENO UKAERE
WELL NAME - OCSG BILABRI DX1	LOCATION BILABRI	COUNTY-PARISH-BLOCK	
DISTRICT NIGERIA	DRILLING CONTRACTOR RIG # BULTO RD DOLPHIN	TYPE OF JOB 20" CASING CMT.	

SIZE & TYPE OF PLUGS		LIST-CSG-HARDWARE		PHYSICAL SLURRY PROPERTIES				
TOP				SLURRY WGT PPG	SLURRY YLD FT ³	WATER GPS	PUMP TIME HR:MIN	Bbl SLURRY
BTM								Bbl MIX WATER
MATERIALS FURNISHED BY BJ								
CEMENT AND ADDITIVES				12	2.517	14.6	1:49	405
				15.8	1.1	5.05	1:04	152

Available Mix Water _____ Bbl.	Available Displ. Fluid _____ Bbl.	TOTAL	557	500
--------------------------------	-----------------------------------	-------	------------	------------

HOLE			TBG-CSG-D.P.			TBG-CSG-D. P.			COLLAR DEPTHS		
SIZE	% EXCESS	DEPTH	SIZE	WGT.	TYPE	DEPTH	SIZE	WGT.	TYPE	DEPTH	SHOE
26	100	348	20"	133#/ft	56	438					
LAST CASING			PKR-CMT RET-BR PL-LINER			PERF. DEPTH			TOP CONN		
SIZE	WGT.	TYPE	DEPTH	BRAND & TYPE		DEPTH	TOP	BTM	SIZE	THREAD	WELL FLUID
CAL. DISPL. VOL. - Bbl			CAL. PSI	CAL. MAX PSI	OP. MAX	MAX TBG PSI			MAX CSG	DISPL. FLUID	
TBG.	CSG.	CSG/D.P.	TOTAL	BUMP PLUG	TO REV.	SQ. PSI	RATED	OP.	DEPTH	TYPE	WGT.

COMMENTS: TROUBLE SETTING TOOL, RUNNING CSG, ETC. PRIOR TO CEMENTING: _____

PRESSURE/RATE DETAIL						JOB EVENT SUMMARY					
TIME HR:MIN.	PRESSURE-PSI		RATE BPM	Bbl. FLUID PUMPED	FLUID TYPE	SAFETY MEETING: BJ CREW ☐ CO. REP. ☐					
	PIPE	ANNULUS				TEST LINES _____ PSI					
						CIRCULATING WELL RIG ☐ BJ ☐					
22	100		4	6	dyed Fluid						
07:26	1000				dyed Fluid	Held For 10 min.					
08:06	300		7	405	slurry	12ppg LEAD SLURRY					
09:55	290		3	152	slurry	15.8ppg TAIL SLURRY					
10:59	320		4	51	sea H ₂ O	DISPLACEMENT					
BUMPED PLUG	PSI TO BUMP PLUG	TEST FLOAT EQUIP.	BARRELS CEMENT RETURNED	TOTAL Bbl. PUMPED	PSI LEFT ON CSG	CAL. TOP OF CEMENT	SERV. SUPV.				
Y N		Y N									



CEMENT JOB REPORT

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CUSTOMER : PEAK/EQUATOR			DATE : 13/01/06		F.R. #		SERV. SUPV. : EWHIMONIH SIMEO					
LEASE & WELL NAME - OCSG			LOCATION : BILABRI				COUNTY-PARISH-BLOCK					
DISTRICT : NIGERIA			DRILLING CONTRACTOR RIG # BULFORD DOLPHIN				TYPE OF JOB : 9 5/8" CSG CMT					
SIZE & TYPE OF PLUGS			LIST-CSG-HARDWARE				PHYSICAL SLURRY PROPERTIES					
TOP	13 3/8" ANTI ROT	7 CENTRALIZERS			SLURRY WGT PPG	SLURRY YLD FT ³	WATER GPS	PUMP TIME HR:MIN	Bbl SLURRY			
BTM	13 3/8" ANTI ROT	1 EACH FLOAT SHOE AND COLLAR										
MATERIALS FURNISHED BY BJ												
G CEMENT, PUMP UNIT, CHEMICAL ADDITIVES & PERSONNEL					15	1.73	7.88	6.40	160			
Available Mix Water <u>97</u> Bbl.					Available Displ. Fluid <u> </u> Bbl.		TOTAL			160		
HOLE			CSG			D. P.			COLLAR DEP			
SIZE	% EXCESS	DEPTH	SIZE	WGT.	TYPE	DEPTH	SIZE	WGT.	TYPE	DEPTH	SHOE	FLOAT
12.25"	20	2779M	9 5/8"	47#/FT		2770M	5"	19.5		177M	2770M	2742
LAST CASING			PKR-CMT RET-BR PL-LINER			PERF. DEPTH			TOP CONN			WE
SIZE	WGT.	TYPE	DEPTH	BRAND & TYPE		DEPTH	TOP	BTM	SIZE	THREAD	TYPE	
13 3/8"	94#/FT		1529M								OBM	
CAL. DISPL. VOL. - Bbl				CAL PSI	CAL MAX PSI	OP. MAX	MAX TBG PSI		MAX CSG	DISPL. FLUID		
TBG.	CSG.	D.P.	TOTAL	BUMP PLUG	TO REV.	SQ. PSI	RATED	OP.	DEPTH	TYPE	WGT.	
	616	10.4	626.4	2000						OBM	10.7PP	
COMMENTS: TROUBLE SETTING TOOL, RUNNING CSG, ETC. PRIOR TO CEMENTING:												
PRESSURE/RATE DETAIL						JOB EVENT SUMMARY						
TIME HR	PRESSURE-PSI		RATE BPM	Bbl. FLUID PUMPED	FLUID TYPE	SAFETY MEETING: BJ CREW ☒ CO. REP. ☒						
	PIPE	ANNULUS				TEST LINES 4000 PSI						
						CIRCULATING WELL RIG ☒ BJ ☐						
03 38	45		3	5	SEA W	FLUSHED LINE.						
03 50	4000				SEA W	TESTED LINE.						
04 03			10	70	SPACER	RIG PUMPED 13 PPG SPACER AND DROPPED BOTTOM PLUG.						
05 28	200		4.5	160	SLURRY	MIXED AND PUMPED 160BBL 15PPG CLASS G PLUS SILICA FLO						
						CMT SLURRY.						
05 38	200		5	20	SEA W	DROPPED TOP PLUG AND KICKED IT OFF WITH 15 BBL OBM.						
05 39			8	606.4		HANDLED OVER DIPLACEMENT OF 606.4 BBL TO RIG						
BUMPED PLUG	PSI TO BUMP PLUG	TEST FLOAT EQUIP.	BARRELS CEMENT RETURNED	TOTAL Bbl. PUMPED	PSI LEFT ON CSG	CAL. TOP OF CEMENT	SERV. SUPV. EWHIMONIH SIMEON					
YES	2000	YES		861.4	5							

P 0



CEMENT JOB REPORT

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TOMER : PEAK/EQUATOR			DATE : 27/01/06			F.R. #			SERV. SUPV. : KENE UKAERE				
LEASE & WELL NAME - OCSG BILABRI DX1			LOCATION : BILABRI						COUNTY-PARISH-BLOCK				
DISTRICT : NIGERIA			DRILLING CONTRACTOR RIG # BULFORD DOLPHIN						TYPE OF JOB : 7" LINER CMT				
SIZE & TYPE OF PLUGS			LIST-CSG-HARDWARE			PHYSICAL SLURRY PROPERTIES							
TOP			4CENTRALIZERS			SLURRY WGT PPG	SLURRY YLD FT ³	WATER GPS	PUMP TIME HR:MIN	Bbl SLURRY	Bbl MIX WATER		
BTM			1 EACH FLOAT SHOE AND COLLAR										
MATERIALS FURNISHED BY BJ													
G CEMENT, PUMP UNIT, CHEMICAL ADDITIVES & PERSONNEL						15.8	1.54	3.91	0.40	37	21		
able Mix Water 97 Bbl.						Available Displ. Fluid Bbl.			TOTAL 37 21				
HOLE			CSG			D. P.			COLLAR DEPTHS				
SIZE	% EXCESS	DEPTH	SIZE	WGT.	TYPE	DEPTH	SIZE	WGT.	TYPE	DEPTH	SHOE	FLOAT	STAGE
81/2"	20	M	7"	29#/FT		3079M	5"	19.5		2716M	3079M	3052.03M	1
LAST CASING			PKR-CMT RET-BR PL-LINER			PERF. DEPTH			TOP CONN			WELL FLUID	
SIZE	WGT.	TYPE	DEPTH	BRAND & TYPE		DEPTH	TOP	BTM	SIZE	THREAD	TYPE	WGT.	
95/8"	47#/FT		2770M								OBM	13.2 PPG	
CAL. DISPL. VOL. - Bbl				CAL PSI	CAL MAX PSI	OP. MAX	MAX TBG PSI		MAX CSG	DISPL. FLUID		WATER SOURCE	
TBG.	CSG.	D.P.	TOTAL	BUMP PLUG	TO REV.	SQ. PSI	RATED	OP.	DEPTH	TYPE	WGT.		
	43	149	192	800						OBM	13.2 PPG	SEA	
COMMENTS: TROUBLE SETTING TOOL, RUNNING CSG, ETC. PRIOR TO CEMENTING:													
PRESSURE/RATE DETAIL						JOB EVENT SUMMARY							
TIME HR	PRESSURE-PSI		RATE BPM	Bbl. FLUID PUMPED	FLUID TYPE	SAFETY MEETING: BJ CREW ☒ CO. REP. ☒							
	PIPE	ANNULUS				TEST LINES 4000 PSI							
						CIRCULATING WELL RIG ☒ BJ ☐							
07 40	100		3	5	SEA W	FLUSHED LINE.							
07 47	3000				SEA W	TESTED LINE.							
07 50			10	60	SPACER	RIG PUMPED 14.2 PPG SPACER AND DROPPED BOTTOM PLUG.							
08 28	200		3	37	SLURRY	MIXED AND PUMPED 37BBL 15.8PPG CLASS G PLUS SILICA FLOWER							
						CMT SLURRY.							
09 12	800		6	192	OBM	DISPLACEMENT							
BUMPED PLUG YES	PSI TO BUMP PLUG 800	TEST FLOAT EQUIP. YES	BARRELS CEMENT RETURNED	TOTAL BH. PUMPED 294	PSI LEFT ON CSG 5	CAL. TOP OF CEMENT	SERV. SUPV. KENE UKAERE						



CEMENT JOB REPORT

Page 1 of 1

CUSTOMER : PEAK/EQUATOR	DATE : 18/02/06	F.R. #	SERV. SUPV. : EWHIMONIH SIMEON
LEASE & WELL NAME - OCSG	LOCATION : BILABRI	COUNTY-PARISH-BLOCK	
DISTRICT : NIGERIA	DRILLING CONTRACTOR RIG # BULFORD DOLPHIN	TYPE OF JOB : BULLHEAD CMT	

SIZE & TYPE OF PLUGS		LIST-CSG-HARDWARE	PHYSICAL SLURRY PROPERTIES					
TOP	13 3/8" ANTI ROT	7 CENTRALIZERS	SLURRY WGT PPG	SLURRY YLD FT ³	WATER GPS	PUMP TIME HR:MIN	Bbl SLURRY	Bbl MIX WATER
BTM	13 3/8" ANTI ROT	1 EACH FLOAT SHOE AND COLLAR						

MATERIALS FURNISHED BY BJ

G CEMENT, PUMP UNIT, CHEMICAL ADDITIVES & PERSONNEL

15.8 1.54 6.45 5 117 66

Available Mix Water <u>66</u> Bbl.	Available Displ. Fluid <u> </u> Bbl.	TOTAL	117	66
------------------------------------	---	-------	-----	----

HOLE			CSG			D. P.			COLLAR DEPTHS				
SIZE	% EXCESS	DEPTH	SIZE	WGT.	TYPE	DEPTH	SIZE	WGT.	TYPE	DEPTH	SHOE	FLOAT	STAGE
6"		2941M	7"	29#/FT		2941M	5"	19.5		177M	2770M		1

LAST CASING				PKR-CMT RET-BR PL-LINER		PERF. DEPTH		TOP CONN		WELL FLUID		
SIZE	WGT.	TYPE	DEPTH	BRAND & TYPE		DEPTH	TOP	BTM	SIZE	THREAD	TYPE	WGT.
											OBM	15PPG

CAL. DISPL. VOL. - Bbl				CAL PSI	CAL MAX PSI	OP. MAX	MAX TBG PSI	MAX CSG	DISPL. FLUID		WATER SOURCE	
TBG.	CSG.	D.P.	TOTAL	BUMP PLUG	TO REV	SQ. PSI	RATED	OP.	DEPTH	TYPE	WGT.	
										OBM	15PPG	SEA

COMMENTS: TROUBLE SETTING TOOL, RUNNING CSG, ETC. PRIOR TO CEMENTING:

PRESSURE/RATE DETAIL						JOB EVENT SUMMARY	
TIME HR	PRESSURE-PSI		RATE BPM	Bbl. FLUID PUMPED	FLUID TYPE	SAFETY MEETING: BJ CREW ☒ CO. REP. ☒	
	PIPE	ANNULUS				TEST LINES 4000 PSI	CIRCULATING WELL RIG ☒ BJ ☐
5 41	1500		2.5	10	DRILL W	PUMP DRILL WATER AHEAD.	
15 47	3500				DRILL W	TESTED LINE.	
15 55	1500		2.5	20	DRILL W	PUMP DRILL WATER AHEAD.	
16 54	2000		4	117	SLURRY	MIXED AND PUMPED 117BBL 15.8PPG CLASS G PLUS SILICA FLOWER	
						CMT SLURRY.	
17 38	2000		4	140	DISPL F	DISPLACE CMT WITH 15 PPG OBM AFTER PUMPING DRILL WATER BEHIND.	
02 02	1000		1	6	DRILL W	TESTED CMT FOR 30 MINUTES OKAY	
BUMPED PLUG	PSI TO BUMP PLUG	TEST FLOAT EQUIP.	BARRELS CEMENT RETURNED	TOTAL BBL PUMPED	PSI LEFT ON CSG	CAL. TOP OF CEMENT	SERV. SUPV. EWHIMONIH SIMEON
YES	2000	YES		293	550	3000M	<i>[Signature]</i>

Checked by Co. Rep.



BJ Services Company Ltd.

Client: Peak Petroleum & Equator Exploration Nigeria Limited

Project Name: Offshore Nigeria Block OML 122

Well Name: Bilabri D1

BJ Services Company Ltd.

WELL COST

Actual x Planned



BJ Services Company Ltd.
Client: Peak Petroleum & Equator Exploration Nigeria Limited
Project Name: Offshore Nigeria Block OML 122
Well Name: Bilabri D1

COST ANALYSIS FOR Bilabri D1

PLANNED x FINAL

	PLANNED USD	FINAL USD
<u>20" & 30" CASING</u>	74,395.04	126,154.00
<u>13 3/8" CASING</u>	115,027.80	171,968.50
<u>9 5/8" CASING</u>	303,505.70	392,996.65
<u>7" LINER</u>	96,699.82	54,830.20
TOTAL	589,628.36	745949.35

NOTE:

- ❖ The difference between the planned and the final well cost occurred due to losses encountered during the job and also to the rig pits dead volume where the cement slurry or spacer was prepared.
- ❖ The well cost excludes personnel and equipment cost.



BJ Services Company Ltd.
Client: Peak Petroleum & Equator Exploration Nigeria Limited
Project Name: Offshore Nigeria Block OML 122
Well Name: Bilabri D1
BJ Services Company Ltd.

JOB EVALUATION REPORTS

CUSTOMER JOB EVALUATION

Indicator	Performance	
1. Did we pump the job as designated/requested?	☒ Yes	☐ No
2. Were the proper prejob and postjob QC tests performed?	☒ Yes	☐ No
3. Were we ready to pump when specified?	☒ Yes	☐ No
4. Did we conduct a thorough and timely prejob and postjob review with you?	☒ Yes	☐ No
5. Do you feel this job was performed in a safe manner?	☒ Yes	☐ No
6. Was the performance and appearance of our personnel professional?	☒ Yes	☐ No
7. Did our products satisfy your requirements?	☒ Yes	☐ No
8. Were you satisfied with our equipment performance and appearance?	☒ Yes	☐ No
Comments/What can we do to improve the next Job?		
<i>Good Job</i>		
<i>Hard working crew</i>		
<i>Before</i>		
Would you like someone from our company to contact you regarding any of your concerns?	☐ Yes	☒ No

Quality
Improvement



Job Evaluation

DATE: 21/12/05
Customer: PEAK/EQUATOR
Well Name: BILABRI DX1
Rig: BULFORD DOLPHIN
Job Supervisor: SIMEON EWHIMONIH
Customer Rep: TEXAS RICHARDS
Signature: <i>[Signature]</i>

ATTN:

SAM ELENON

QUALITY & ASSURANCE

CUSTOMER JOB EVALUATION

Quality Improvement



Job Evaluation

Indicator	Performance	
1. Did we pump the job as designated/requested?	Yes ☒	No ☐
2. Were the proper prejob and postjob QC tests performed?	Yes ☒	No ☐
3. Were we ready to pump when specified?	Yes ☒	No ☐
4. Did we conduct a thorough and timely prejob and <u>postjob</u> review with you?	Yes ☒	No ☐
5. Do you feel this job was performed in a safe manner?	Yes ☒	No ☐
6. Was the performance and appearance of our personnel professional?	Yes ☒	No ☐
7. Did our products satisfy your requirements?	Yes ☒	No ☐
8. Were you satisfied with our equipment performance and appearance?	Yes ☐	No ☒
Comments/What can we do to improve the next Job?		
<i>I don't think the LAS system</i>		
<i>is as efficient, when not using</i>		
<i>the pits for mixing more than</i>		
<i>4 additives.</i>		
<i>No more crew changes, on</i>		
<i>the day of a cement job!</i>		
Would you like someone from our company to contact you regarding any of your concerns?	Yes ☐	No ☐

DATE: 13/01/06

Customer: PEAK/EQUATOR

Well Name: BILABRI DX1

Rig: BULFORD DOLPHIN

Job Supervisor: SIMEON EWHIMONIH

Customer Rep: TEXAS RICHARDS

Signature:

Texas

CUSTOMER JOB EVALUATION

Indicator	Performance	
1. Did we pump the job as designated/requested?	Yes	☒ No
2. Were the proper prejob and postjob QC tests performed?	☒ Yes	No
3. Were we ready to pump when specified?	☒ Yes	No
4. Did we conduct a thorough and timely prejob and postjob review with you?	☒ Yes	No
5. Do you feel this job was performed in a safe manner?	☒ Yes	No
6. Was the performance and appearance of our personnel professional?	☒ Yes	No
7. Did our products satisfy your requirements?	☒ Yes	No
8. Were you satisfied with our equipment performance and appearance?	Yes	☒ No
Comments/What can we do to improve the next Job?		
Danhole rate was too		
low, later investigation		
found blockage in mixing		
system & faulty valve to		
surge can		
Would you like someone from our company to contact you regarding any of your concerns?	Yes	No

Quality Improvement



30" CASING Job Evaluation

DATE:	28-11-05
Customer:	PEAK/EQUATOR PET.
Well Name:	BILABRI DX1
Rig:	BUFOLD DOLPHIN
Job Supervisor:	KENE/SIMEON
Customer Name:	S GREEN
Signature	

CUSTOMER JOB EVALUATION

Indicator	Performance	
1. Did we pump the job as designated/requested?	Yes	No
2. Were the proper prejob and postjob QC tests performed?	Yes	No
3. Were we ready to pump when specified?	Yes	No
4. Did we conduct a thorough and timely prejob and postjob review with you?	Yes	No
5. Do you feel this job was performed in a safe manner?	Yes	No
6. Was the performance and appearance of our personnel professional?	Yes	No
7. Did our products satisfy your requirements?	Yes	No
8. Were you satisfied with our equipment performance and appearance?	Yes	No
Comments/What can we do to improve the next Job?		
Would you like someone from our company to contact you regarding any of your concerns?	Yes	No

**Quality
Improvement**



DATE: 18-20/03/06

Customer: PEAK/EQUATOR

Well Name: BILABRI DX1

Rig: BULFORD DOLPHIN

Job Supervisor: SIMEON EWHIMONIH

Customer Rep: TEXAS RICHARDS

Signature:

CUSTOMER JOB EVALUATION

Indicator	Performance	
1. Did we pump the job as designated/requested?	Yes	☒ No
2. Were the proper prejob and postjob QC tests performed?	☒ Yes	No
3. Were we ready to pump when specified?	☒ Yes	No
4. Did we conduct a thorough and timely prejob and postjob review with you?	☒ Yes	No
5. Do you feel this job was performed in a safe manner?	☒ Yes	No
6. Was the performance and appearance of our personnel professional?	☒ Yes	No
7. Did our products satisfy your requirements?	☒ Yes	No
8. Were you satisfied with our equipment performance and appearance?	Yes	☒ No
Comments/What can we do to improve the next Job?		
Unable to obtain designed bail		
& lead slurry w/ls due to		
limitations and blockages in		
the mixing equipment,		
Would you like someone from our company to contact you regarding any of your concerns?	☒ Yes	No

Quality Improvement



20" CASING Job Evaluation

DATE: 07-12-05
Customer: PEAK / EQUATOR
Well Name: BILABRI DX1
Rig: BULFORD DOLPHIN
Job Supervisor: KENE UKAERC
Customer Name: S. GREEN / PEAK
Signature:  8-12-05